

## read\_predictions\_baseline\_top\_1

February 5, 2026

```
[161]: import pandas as pd
import json
import re
from sklearn.metrics import classification_report, confusion_matrix, \
    accuracy_score
```

```
[162]: run_name = "baseline"
run_name = "baseline_lh"
run_name = "baseline_lh_2"
run_name = "baseline_lh_3"

run_name = "baseline_lh_1"
run_name = "baseline_lh_0"
run_name = "baseline"

# load JSONL
records = []
with open("../aircraft_er_predictions/" + run_name + "_predictions_all.tsv") as f:
    for line in f:
        records.append(json.loads(line))

# load JSONL
records_test = []
with open("../aircraft_er_predictions/" + run_name + "_predictions_test.tsv") as f:
    for line in f:
        records_test.append(json.loads(line))
```

```
[163]: df_all = pd.DataFrame(records)
df_test = pd.DataFrame(records_test)
df_test.head()
```

```
[163]:                                left \
0  COL make VAL BOEING COL model VAL 707 COL seri...
1  COL make VAL FAIRCHILD COL model VAL PILGRIM 1...
2  COL make VAL AEROSPATIALE COL model VAL SA330 ...
```

```

3 COL make VAL BOEING COL model VAL 727 COL seri...
4 COL make VAL PILATUS COL model VAL PC6 COL ser...

```

|   |                                                   | right | match | match_confidence |
|---|---------------------------------------------------|-------|-------|------------------|
| 0 | COL make VAL PC COL model VAL 6 COL series VAL... | 0     |       | 0.999997         |
| 1 | COL make VAL THNDCT COL model VAL COLT COL ser... | 0     |       | 0.999996         |
| 2 | COL make VAL SA COL model VAL 330 COL series V... | 1     |       | 0.999919         |
| 3 | COL make VAL B COL model VAL 727 COL series VA... | 0     |       | 0.999984         |
| 4 | COL make VAL PC COL model VAL 6 COL series VAL... | 1     |       | 0.999897         |

```

[164]: gold_test = pd.read_csv("../data/ditto_aircraft/" + run_name + "/test.txt",
    ↪sep="\t", header=None, names=["left", "right", "gold"])

gold_all = pd.read_csv("../data/ditto_aircraft/" + run_name + "/all_pairs.txt",
    ↪sep="\t", header=None, names=["left", "right", "gold"])
print(gold_all.head())

```

|   |                                                   | left | \ |
|---|---------------------------------------------------|------|---|
| 0 | COL make VAL GULFSTREAM COL model VAL GVI COL ... |      |   |
| 1 | COL make VAL PIAGGIO COL model VAL P136 COL se... |      |   |
| 2 | COL make VAL AERONCA COL model VAL L3 COL seri... |      |   |
| 3 | COL make VAL PZL OKECIE COL model VAL PZL104 C... |      |   |
| 4 | COL make VAL BEECH COL model VAL 99 COL series... |      |   |

  

|   |                                                   | right | gold |
|---|---------------------------------------------------|-------|------|
| 0 | COL make VAL PA COL model VAL 46 COL series VA... | 0     |      |
| 1 | COL make VAL BHT COL model VAL UH1E COL series... | 0     |      |
| 2 | COL make VAL LR COL model VAL 45 COL series VA... | 0     |      |
| 3 | COL make VAL B COL model VAL 737 COL series VA... | 0     |      |
| 4 | COL make VAL BE COL model VAL 18 COL series VA... | 0     |      |

```

[165]: df_all["gold"] = gold_all["gold"]
df_test["gold"] = gold_test["gold"]

```

```

[166]: #Test Data

y_true_test = df_test["gold"]
y_pred_test = df_test["match"]

print("Accuracy:", accuracy_score(y_true_test, y_pred_test))
print("\nClassification report:\n", classification_report(y_true_test,
    ↪y_pred_test, digits=3))
print("\nConfusion matrix:\n", confusion_matrix(y_true_test, y_pred_test))

```

```

with open("../aircraft_er_predictions/" + run_name + "_metrics_test.txt", "w") as f:
    print("Accuracy:", accuracy_score(y_true_test, y_pred_test), file=f)
    print("\nClassification report:\n", file=f)
    print(classification_report(y_true_test, y_pred_test, digits=3), file=f)
    print("\nConfusion matrix:\n", file=f)
    print(confusion_matrix(y_true_test, y_pred_test), file=f)

```

Accuracy: 0.9980222496909765

Classification report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.999     | 0.998  | 0.999    | 3677    |
| 1            | 0.984     | 0.995  | 0.989    | 368     |
| accuracy     |           |        | 0.998    | 4045    |
| macro avg    | 0.992     | 0.996  | 0.994    | 4045    |
| weighted avg | 0.998     | 0.998  | 0.998    | 4045    |

Confusion matrix:

```

[[3671   6]
 [   2 366]]

```

[167]: #All Data

```

y_true = df_all["gold"]
y_pred = df_all["match"]

print("Accuracy:", accuracy_score(y_true, y_pred))
print("\nClassification report:\n", classification_report(y_true,
    y_pred, digits=3))
print("\nConfusion matrix:\n", confusion_matrix(y_true, y_pred))

with open("../aircraft_er_predictions/" + run_name + "_metrics_all.txt", "w") as f:
    print("Accuracy:", accuracy_score(y_true, y_pred), file=f)
    print("\nClassification report:\n", file=f)
    print(classification_report(y_true, y_pred, digits=3), file=f)
    print("\nConfusion matrix:\n", file=f)
    print(confusion_matrix(y_true, y_pred), file=f)

```

Accuracy: 0.9993067247697336

Classification report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 1.000     | 1.000  | 1.000    | 36710   |
| 1            | 0.996     | 0.997  | 0.996    | 3678    |
| accuracy     |           |        | 0.999    | 40388   |
| macro avg    | 0.998     | 0.998  | 0.998    | 40388   |
| weighted avg | 0.999     | 0.999  | 0.999    | 40388   |

Confusion matrix:

```
[[36694  16]
 [  12 3666]]
```

```
[168]: errors = df_test[df_test["gold"] != df_test["match"]]
print(errors[["left", "right", "gold", "match", "match_confidence"]].head(20))
```

|      | left \                                            | right | gold | match \ |
|------|---------------------------------------------------|-------|------|---------|
| 636  | COL make VAL LEARJET COL model VAL 35 COL seri... |       |      |         |
| 1102 | COL make VAL BEECH COL model VAL AT10 COL seri... |       |      |         |
| 1713 | COL make VAL MITSUBISHI COL model VAL MU2B COL... |       |      |         |
| 1845 | COL make VAL BELL COL model VAL UH1 COL series... |       |      |         |
| 2152 | COL make VAL AERO DIFUSION COL model VAL D1190... |       |      |         |
| 2962 | COL make VAL CESSNA COL model VAL 402 COL seri... |       |      |         |
| 3615 | COL make VAL REARWIN COL model VAL 8500 COL se... |       |      |         |
| 3899 | COL make VAL HAWKER SIDDELEY COL model VAL HS1... |       |      |         |
|      |                                                   | right | gold | match \ |
| 636  | COL make VAL LR COL model VAL 35 COL series VA... | 1     | 0    |         |
| 1102 | COL make VAL VERVLE COL model VAL AT COL serie... | 0     | 1    |         |
| 1713 | COL make VAL MU COL model VAL 2B COL series VA... | 0     | 1    |         |
| 1845 | COL make VAL FEDRIC COL model VAL UH34 COL ser... | 0     | 1    |         |
| 2152 | COL make VAL AMD COL model VAL 50 COL series V... | 0     | 1    |         |
| 2962 | COL make VAL CE COL model VAL 401 COL series V... | 0     | 1    |         |
| 3615 | COL make VAL COMWTH COL model VAL 9000 COL ser... | 1     | 0    |         |
| 3899 | COL make VAL HS COL model VAL 125 COL series V... | 0     | 1    |         |
|      | match_confidence                                  |       |      |         |
| 636  | 0.999731                                          |       |      |         |
| 1102 | 0.998443                                          |       |      |         |
| 1713 | 0.999841                                          |       |      |         |
| 1845 | 0.870332                                          |       |      |         |
| 2152 | 0.792658                                          |       |      |         |
| 2962 | 0.999301                                          |       |      |         |
| 3615 | 0.999971                                          |       |      |         |
| 3899 | 0.558602                                          |       |      |         |

```
[169]: errors
```

```
[169]:
```

|      | left \                                            | right match \ |
|------|---------------------------------------------------|---------------|
| 636  | COL make VAL LEARJET COL model VAL 35 COL seri... | 0             |
| 1102 | COL make VAL BEECH COL model VAL AT10 COL seri... | 1             |
| 1713 | COL make VAL MITSUBISHI COL model VAL MU2B COL... | 1             |
| 1845 | COL make VAL BELL COL model VAL UH1 COL series... | 1             |
| 2152 | COL make VAL AERO DIFUSION COL model VAL D1190... | 1             |
| 2962 | COL make VAL CESSNA COL model VAL 402 COL seri... | 1             |
| 3615 | COL make VAL REARWIN COL model VAL 8500 COL se... | 0             |
| 3899 | COL make VAL HAWKER SIDDELEY COL model VAL HS1... | 1             |

  

|      | match_confidence | gold |
|------|------------------|------|
| 636  | 0.999731         | 1    |
| 1102 | 0.998443         | 0    |
| 1713 | 0.999841         | 0    |
| 1845 | 0.870332         | 0    |
| 2152 | 0.792658         | 0    |
| 2962 | 0.999301         | 0    |
| 3615 | 0.999971         | 1    |
| 3899 | 0.558602         | 0    |

```
[170]: errors.to_csv("../aircraft_er_predictions/" + run_name + "_errors_review.csv",
    ↪index=False)
```

```
[171]: def parse_record(record: str):
    """Parse Ditto serialized record into a dict of {field: value}."""
    parts = re.split(r"COL |VAL ", record.strip())
    parts = [p for p in parts if p] # drop empties
    return {parts[i].strip(): parts[i+1].strip() for i in range(0, len(parts),
    ↪2)}

parsed = []

for _, row in errors.iterrows():
    left = parse_record(row["left"])
    right = parse_record(row["right"])
    parsed.append({
        "cicitt_make": left.get("make"),
```

```

    "make": right.get("make"),
    "cictt_model": left.get("model"),
    "model": right.get("model"),
    "cictt_series": left.get("series"),
    "series": right.get("series"),
    "cictt_cert": left.get("cert"),
    "cert": right.get("cert"),
    "cictt_name": left.get("name"),
    "name": right.get("name"),
    "predicted": row["match"],
    "confidence": row["match_confidence"],
    "gold": row["gold"]
})

```

```
aligned = pd.DataFrame(parsed)
```

```
[172]: aligned.to_csv("../aircraft_er_predictions/" + run_name +
    ↪ "_aligned_errors_review.csv", index=False)
```

```
[173]: aligned.head()
```

```
[173]:
```

|   | cictt_make    | make   | cictt_model | model | cictt_series | series  | \ |
|---|---------------|--------|-------------|-------|--------------|---------|---|
| 0 | LEARJET       | LR     | 35          | 35    | A            | 35AC21A |   |
| 1 | BEECH         | VERVLE | AT10        | AT    | AT10         | AT      |   |
| 2 | MITSUBISHI    | MU     | MU2B        | 2B    | 26A          | 20      |   |
| 3 | BELL          | FEDRIC | UH1         | UH34  | E            | E       |   |
| 4 | AERO DIFUSION | AMD    | D1190       | 50    | S            | 900     |   |

  

|   | cictt_cert       | cert   | cictt_name      | name           | predicted | \ |
|---|------------------|--------|-----------------|----------------|-----------|---|
| 0 | A10CE            | A10CE  | No Popular Name | None           | 0         |   |
| 1 | LTC-12           | ATC323 | WICHITA         | None           | 1         |   |
| 2 | NOT IDENTIFIED   | A2PC   | MARQUISE        | None           | 1         |   |
| 3 | H7SO             | H7NM   | IROQUOIS        | None           | 1         |   |
| 4 | NOT CERTIFICATED | A46EU  | COMPOSTELA      | MYSTERE-FALCON | 1         |   |

  

|   | confidence | gold |
|---|------------|------|
| 0 | 0.999731   | 1    |
| 1 | 0.998443   | 0    |
| 2 | 0.999841   | 0    |
| 3 | 0.870332   | 0    |
| 4 | 0.792658   | 0    |

### 0.0.1 Top 1 Approach

Start with correct intuition. Understand the correct answer per row. Need the id as a primary key

```
[174]: # This has the left_ids and right_ids included
pairs_with_id_path = f"../data/ditto_aircraft/{run_name}/all_pairs_with_id.txt"
pairs_with_id = pd.read_csv(pairs_with_id_path)
```

```
df_all["left_id"] = pairs_with_id["left_id"]
df_all["right_id"] = pairs_with_id["right_id"]
```

```
[175]: #Add primary key left_id and right_id from pairs_with_id file
```

```
df_all["left_id"] = pairs_with_id["left_id"]
df_all["right_id"] = pairs_with_id["right_id"]
```

```
#Analyze result for each row determining TPs, FPs, etc.
```

```
df_all["TP"] = ((df_all["match"] == 1) & (df_all["gold"] == 1)).astype(int)
df_all["TN"] = ((df_all["match"] == 0) & (df_all["gold"] == 0)).astype(int)
df_all["FP"] = ((df_all["match"] == 1) & (df_all["gold"] == 0)).astype(int)
df_all["FN"] = ((df_all["match"] == 0) & (df_all["gold"] == 1)).astype(int)
```

```
[176]: df_all.head()
```

```
[176]:
```

```
left \
0 COL make VAL GULFSTREAM COL model VAL GVI COL ...
1 COL make VAL PIAGGIO COL model VAL P136 COL se...
2 COL make VAL AERONCA COL model VAL L3 COL seri...
3 COL make VAL PZL OKECIE COL model VAL PZL104 C...
4 COL make VAL BEECH COL model VAL 99 COL series...
```

```
right match match_confidence \
0 COL make VAL PA COL model VAL 46 COL series VA... 0 0.999998
1 COL make VAL BHT COL model VAL UH1E COL series... 0 0.999994
2 COL make VAL LR COL model VAL 45 COL series VA... 0 0.999996
3 COL make VAL B COL model VAL 737 COL series VA... 0 0.999995
4 COL make VAL BE COL model VAL 18 COL series VA... 0 0.999997
```

```
gold left_id right_id TP TN FP FN
0 0 10257 2641 0 1 0 0
1 0 7344 586 0 1 0 0
2 0 165 2223 0 1 0 0
3 0 7824 876 0 1 0 0
4 0 1318 345 0 1 0 0
```

```
[177]: #For ech right_id, sum TPs, FPs, etc by right id
```

```
cols = ["TP", "FP", "TN", "FN"]
```

```
by_right = (
    df_all
```

```

        .groupby("right_id", as_index=False)[cols]
        .sum()
    )

by_right

```

```

[177]:
   right_id  TP  FP  TN  FN
0         1   1   0   7   0
1         2   1   0  12   0
2         3   1   0  14   0
3         4   1   0  12   0
4         5   1   0   9   0
...
3673    3735   1   0   8   0
3674    3736   1   0  10   0
3675    3737   1   0  10   0
3676    3738   1   0   8   0
3677    3739   1   0   9   0

```

[3678 rows x 5 columns]

```

[178]: #This is a good start on the confusion matrix but it doesn't tie break by
       ↳ confidence
df_all_merged[cols].sum()

```

```

[178]: TP      3666
       FP       16
       TN    36694
       FN       12
       dtype: int64

```

```

[179]: # Agregating by right_id shows stats for each right_id
df_agg_right = (
    df_all.groupby('right_id', as_index=False)
        .agg({'right':      'first',
              'TP':        'sum',
              'TN':        'sum',
              'FP':        'sum',
              'FN':        'sum',
              'left':      lambda s: pd.unique(s.dropna()).tolist(),
            })
)
df_agg_right["left_len"] = df_agg_right["left"].apply(len)

df_agg_right.head()

```



```
[179]:
```

|   | right_id |                                                   | right | TP | TN | FP | \ |
|---|----------|---------------------------------------------------|-------|----|----|----|---|
| 0 | 1        | COL make VAL AERPEG COL model VAL M100S COL se... | 1     | 7  | 0  |    |   |
| 1 | 2        | COL make VAL AL COL model VAL 60 COL series VA... | 1     | 12 | 0  |    |   |
| 2 | 3        | COL make VAL AL COL model VAL 60 COL series VA... | 1     | 14 | 0  |    |   |
| 3 | 4        | COL make VAL MRCHTI COL model VAL F260 COL ser... | 1     | 12 | 0  |    |   |
| 4 | 5        | COL make VAL MRCHTI COL model VAL F260 COL ser... | 1     | 9  | 0  |    |   |

  

|   | FN |                                                   | left | left_len |
|---|----|---------------------------------------------------|------|----------|
| 0 | 0  | [COL make VAL BEECH COL model VAL SNB COL seri... | 8    |          |
| 1 | 0  | [COL make VAL SCHLEICHER COL model VAL ASW12 C... | 13   |          |
| 2 | 0  | [COL make VAL STAR COL model VAL CAVALIER COL ... | 15   |          |
| 3 | 0  | [COL make VAL AERMACCHI COL model VAL F260 COL... | 13   |          |
| 4 | 0  | [COL make VAL BOEING COL model VAL 737 COL ser... | 10   |          |

```
[180]: # Calculate the maximum match confidence only for classified matches
df_agg_max_conf_right = (
    df_all[df_all["match"]==1].groupby('right_id', as_index=False)
    .agg({'match_confidence': 'max',
        })
)
```

```
[181]: # Merge exhaustive right rows with merged right rows
df_all_merged_right = df_all.merge(
    df_agg_right,
    on="right_id",
    how="left",
    suffixes=("", "_agg"))
```

```
[182]: # Merge max match confidence score
df_all_merged_right = df_all_merged_right.merge(
    df_agg_max_conf_right,
    on="right_id",
    how="left",
    suffixes=("", "_max"))
```

```
[183]: # A best (top 1) match is when there is a match and the match has the maximum
↳ confidence
df_all_merged_right["best"] = ((df_all_merged["match"]==1) &
↳ (df_all_merged["match_confidence"] ==
↳ df_all_merged["match_confidence_max"])).astype(int)
```

```
[184]: #Check that each right_id has one best match
df_all_merged_right[df_all_merged_right["best"] == 1]["right_id"].value_counts()
```

```
[184]: right_id
2316    1
895     1
```

```

2476    1
1264    1
3243    1
..
873     1
143     1
832     1
3133    1
2760    1
Name: count, Length: 3451, dtype: int64

```

```
[185]: #All Data -compare best match (top 1) with gold
```

```

y_true = df_all_merged_right["gold"]
y_pred = df_all_merged_right["best"]

print("Accuracy:", accuracy_score(y_true, y_pred))
print("\nClassification report:\n", classification_report(y_true,
    ↳y_pred,digits=3))
print("\nConfusion matrix:\n", confusion_matrix(y_true, y_pred))

```

Accuracy: 0.9943795186689116

Classification report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.994     | 1.000  | 0.997    | 36710   |
| 1            | 1.000     | 0.938  | 0.968    | 3678    |
| accuracy     |           |        | 0.994    | 40388   |
| macro avg    | 0.997     | 0.969  | 0.983    | 40388   |
| weighted avg | 0.994     | 0.994  | 0.994    | 40388   |

Confusion matrix:

```

[[36710    0]
 [  227 3451]]

```

```
[186]: right_errors = df_all_merged_right[df_all_merged_right["gold"] !=
    ↳df_all_merged_right["best"]]
```

```
[187]: right_errors
```

```

[187]:                                     left \
102    COL make VAL GRUMMAN COL model VAL G21 COL ser...
174    COL make VAL DOUGLAS COL model VAL C53 COL ser...
261    COL make VAL DOUGLAS COL model VAL C53 COL ser...
265    COL make VAL FAIRCHILD COL model VAL C123 COL ...

```

454 COL make VAL SAAB COL model VAL 340 COL series...  
 ...  
 39708 COL make VAL PIPER COL model VAL PA25 COL seri...  
 39815 COL make VAL BOEING COL model VAL 747 COL seri...  
 39877 COL make VAL BELL COL model VAL UH1 COL series...  
 39960 COL make VAL REARWIN COL model VAL 8500 COL se...  
 40077 COL make VAL BELL COL model VAL UH1 COL series...

|       |                                                   | right | match | \ |
|-------|---------------------------------------------------|-------|-------|---|
| 102   | COL make VAL GRUMAN COL model VAL G21 COL seri... |       | 1     |   |
| 174   | COL make VAL DC COL model VAL 3 COL series VAL... |       | 1     |   |
| 261   | COL make VAL DC COL model VAL 3 COL series VAL... |       | 1     |   |
| 265   | COL make VAL REAGAN COL model VAL C123 COL ser... |       | 1     |   |
| 454   | COL make VAL SF COL model VAL 340 COL series V... |       | 1     |   |
| ...   | ...                                               | ...   | ...   |   |
| 39708 | COL make VAL PA COL model VAL 25 COL series VA... |       | 1     |   |
| 39815 | COL make VAL B COL model VAL 747 COL series VA... |       | 1     |   |
| 39877 | COL make VAL RDC COL model VAL UH COL series V... |       | 1     |   |
| 39960 | COL make VAL COMWTH COL model VAL 9000 COL ser... |       | 0     |   |
| 40077 | COL make VAL BHT COL model VAL UH1B COL series... |       | 1     |   |

|       | match_confidence | gold | left_id | right_id | TP  | TN  | FP  | FN  | \ |
|-------|------------------|------|---------|----------|-----|-----|-----|-----|---|
| 102   | 0.999836         | 1    | 5557    | 2057     | 1   | 0   | 0   | 0   |   |
| 174   | 0.910986         | 1    | 16401   | 3436     | 1   | 0   | 0   | 0   |   |
| 261   | 0.978614         | 1    | 16401   | 3439     | 1   | 0   | 0   | 0   |   |
| 265   | 0.999642         | 1    | 21430   | 3534     | 1   | 0   | 0   | 0   |   |
| 454   | 0.999880         | 1    | 8087    | 2741     | 1   | 0   | 0   | 0   |   |
| ...   | ...              | ...  | ...     | ...      | ... | ... | ... | ... |   |
| 39708 | 0.999786         | 1    | 7552    | 2584     | 1   | 0   | 0   | 0   |   |
| 39815 | 0.362639         | 1    | 2609    | 3310     | 1   | 0   | 0   | 0   |   |
| 39877 | 0.999846         | 1    | 1512    | 578      | 1   | 0   | 0   | 0   |   |
| 39960 | 0.999971         | 1    | 7939    | 2699     | 0   | 0   | 0   | 1   |   |
| 40077 | 0.999870         | 1    | 1512    | 582      | 1   | 0   | 0   | 0   |   |

|       |                                                   | right_agg | TP_agg | TN_agg | \ |
|-------|---------------------------------------------------|-----------|--------|--------|---|
| 102   | COL make VAL GRUMAN COL model VAL G21 COL seri... |           | 1      | 10     |   |
| 174   | COL make VAL DC COL model VAL 3 COL series VAL... |           | 1      | 11     |   |
| 261   | COL make VAL DC COL model VAL 3 COL series VAL... |           | 1      | 12     |   |
| 265   | COL make VAL REAGAN COL model VAL C123 COL ser... |           | 1      | 5      |   |
| 454   | COL make VAL SF COL model VAL 340 COL series V... |           | 1      | 15     |   |
| ...   | ...                                               | ...       | ...    | ...    |   |
| 39708 | COL make VAL PA COL model VAL 25 COL series VA... |           | 1      | 7      |   |
| 39815 | COL make VAL B COL model VAL 747 COL series VA... |           | 1      | 9      |   |
| 39877 | COL make VAL RDC COL model VAL UH COL series V... |           | 1      | 8      |   |
| 39960 | COL make VAL COMWTH COL model VAL 9000 COL ser... |           | 0      | 7      |   |
| 40077 | COL make VAL BHT COL model VAL UH1B COL series... |           | 1      | 13     |   |

|       | FP_agg | FN_agg | left_agg \                                        |
|-------|--------|--------|---------------------------------------------------|
| 102   | 0      | 0      | [COL make VAL GRUMMAN COL model VAL G21 COL se... |
| 174   | 0      | 0      | [COL make VAL DOUGLAS COL model VAL C53 COL se... |
| 261   | 0      | 0      | [COL make VAL DOUGLAS COL model VAL C53 COL se... |
| 265   | 0      | 0      | [COL make VAL FAIRCHILD COL model VAL C123 COL... |
| 454   | 0      | 0      | [COL make VAL SAAB COL model VAL 340 COL serie... |
| ...   | ...    | ...    | ...                                               |
| 39708 | 0      | 0      | [COL make VAL AIRBUS COL model VAL A300 COL se... |
| 39815 | 0      | 0      | [COL make VAL SCHLEICHER COL model VAL KA2 COL... |
| 39877 | 0      | 0      | [COL make VAL BOEING COL model VAL 737 COL ser... |
| 39960 | 0      | 1      | [COL make VAL CESSNA COL model VAL 305 COL ser... |
| 40077 | 0      | 0      | [COL make VAL CONVAIR COL model VAL 240 COL se... |

|       | left_len | match_confidence_max | best |
|-------|----------|----------------------|------|
| 102   | 11       | 0.999836             | 0    |
| 174   | 12       | 0.910986             | 0    |
| 261   | 13       | 0.978614             | 0    |
| 265   | 6        | 0.999642             | 0    |
| 454   | 16       | 0.999880             | 0    |
| ...   | ...      | ...                  | ...  |
| 39708 | 8        | 0.999786             | 0    |
| 39815 | 10       | 0.362639             | 0    |
| 39877 | 9        | 0.999846             | 0    |
| 39960 | 8        | NaN                  | 0    |
| 40077 | 14       | 0.999870             | 0    |

[227 rows x 20 columns]

## 0.0.2 Repeat process by left\_id - currently not working

```
[188]: df_agg_cicctt = (
    df_all.groupby('left_id', as_index=False)
    .agg({'left': 'first',
          'TP': 'sum',
          'TN': 'sum',
          'FP': 'sum',
          'FN': 'sum',
          'right': lambda s: pd.unique(s.dropna()).tolist(),
          })
)
df_agg_cicctt["right_len"] = df_agg_cicctt["right"].apply(len)
df_agg_cicctt.head()
```

```
[188]: left_id left TP TN FP FN \
0 25 COL make VAL AER PEGASO COL model VAL M100 COL... 1 11 0 0
```

|   |    |                                                   |   |    |   |   |
|---|----|---------------------------------------------------|---|----|---|---|
| 1 | 46 | COL make VAL AERMACCHI COL model VAL AL60 COL ... | 1 | 13 | 0 | 0 |
| 2 | 47 | COL make VAL AERMACCHI COL model VAL AL60 COL ... | 1 | 10 | 0 | 0 |
| 3 | 48 | COL make VAL AERMACCHI COL model VAL F260 COL ... | 1 | 8  | 0 | 0 |
| 4 | 49 | COL make VAL AERMACCHI COL model VAL F260 COL ... | 1 | 9  | 0 | 0 |

|   |                                                   |  | right | right_len |
|---|---------------------------------------------------|--|-------|-----------|
| 0 | [COL make VAL SCHLER COL model VAL ASW27 COL s... |  |       | 12        |
| 1 | [COL make VAL BB COL model VAL 22 COL series V... |  |       | 14        |
| 2 | [COL make VAL B COL model VAL 737 COL series V... |  |       | 11        |
| 3 | [COL make VAL MRCHTI COL model VAL F260 COL se... |  |       | 9         |
| 4 | [COL make VAL GROB COL model VAL G103 COL seri... |  |       | 10        |

```
[ ]:
```

```
[189]: df_agg_max_conf = (
        df_all[df_all["match"]==1].groupby('left_id', as_index=False)
        .agg({'match_confidence': 'max',
            })
    )
    df_agg_max_conf["max_conf_match"] = df_agg_max_conf["match_confidence"]

    df_agg_max_conf
```

```
[189]:
```

|      | left_id | match_confidence | max_conf_match |
|------|---------|------------------|----------------|
| 0    | 25      | 0.999267         | 0.999267       |
| 1    | 46      | 0.999905         | 0.999905       |
| 2    | 47      | 0.999919         | 0.999919       |
| 3    | 48      | 0.999895         | 0.999895       |
| 4    | 49      | 0.999888         | 0.999888       |
| ...  | ...     | ...              | ...            |
| 3446 | 25445   | 0.999879         | 0.999879       |
| 3447 | 25560   | 0.999897         | 0.999897       |
| 3448 | 25589   | 0.999892         | 0.999892       |
| 3449 | 25626   | 0.999833         | 0.999833       |
| 3450 | 25973   | 0.999743         | 0.999743       |

```
[3451 rows x 3 columns]
```

```
[190]: df_all_merged = df_all.merge(
        df_agg_cicctt,
        on="left_id",
        how="left",
        suffixes=("", "_agg"))
```

```
[191]: df_all_merged = df_all_merged.merge(
        df_agg_max_conf,
        on="left_id",
```

```
how="left",  
suffixes=("", "_max"))
```

```
df_all_merged["best"] = ((df_all_merged["match"]==1) &
    ↳(df_all_merged["match_confidence"] ==
    ↳df_all_merged["match_confidence_max"])).astype(int)
```

```
df_all_merged
```

|       |                                                   |      |         |          |     |     |     |     |   |       |       |   |
|-------|---------------------------------------------------|------|---------|----------|-----|-----|-----|-----|---|-------|-------|---|
|       |                                                   |      |         |          |     |     |     |     |   | left  | \     |   |
| 0     | COL make VAL GULFSTREAM COL model VAL GVI COL ... |      |         |          |     |     |     |     |   |       |       |   |
| 1     | COL make VAL PIAGGIO COL model VAL P136 COL se... |      |         |          |     |     |     |     |   |       |       |   |
| 2     | COL make VAL AERONCA COL model VAL L3 COL seri... |      |         |          |     |     |     |     |   |       |       |   |
| 3     | COL make VAL PZL OKECIE COL model VAL PZL104 C... |      |         |          |     |     |     |     |   |       |       |   |
| 4     | COL make VAL BEECH COL model VAL 99 COL series... |      |         |          |     |     |     |     |   |       |       |   |
| ...   | ...                                               |      |         |          |     |     |     |     |   |       |       |   |
| 40383 | COL make VAL NORTH AMERICAN COL model VAL B25 ... |      |         |          |     |     |     |     |   |       |       |   |
| 40384 | COL make VAL BEECH COL model VAL 17 COL series... |      |         |          |     |     |     |     |   |       |       |   |
| 40385 | COL make VAL CURTISS WRIGHT COL model VAL 22 C... |      |         |          |     |     |     |     |   |       |       |   |
| 40386 | COL make VAL MAULE COL model VAL M6 COL series... |      |         |          |     |     |     |     |   |       |       |   |
| 40387 | COL make VAL HILLER COL model VAL UH12 COL ser... |      |         |          |     |     |     |     |   |       |       |   |
|       |                                                   |      |         |          |     |     |     |     |   |       |       |   |
|       |                                                   |      |         |          |     |     |     |     |   | right | match | \ |
| 0     | COL make VAL PA COL model VAL 46 COL series VA... |      |         |          |     |     |     |     |   | 0     |       |   |
| 1     | COL make VAL BHT COL model VAL UH1E COL series... |      |         |          |     |     |     |     |   | 0     |       |   |
| 2     | COL make VAL LR COL model VAL 45 COL series VA... |      |         |          |     |     |     |     |   | 0     |       |   |
| 3     | COL make VAL B COL model VAL 737 COL series VA... |      |         |          |     |     |     |     |   | 0     |       |   |
| 4     | COL make VAL BE COL model VAL 18 COL series VA... |      |         |          |     |     |     |     |   | 0     |       |   |
| ...   | ...                                               |      |         |          |     |     |     |     |   | ...   |       |   |
| 40383 | COL model VAL B25 COL series VAL J COL cert VA... |      |         |          |     |     |     |     |   | 1     |       |   |
| 40384 | COL make VAL AYRES COL model VAL S2 COL series... |      |         |          |     |     |     |     |   | 0     |       |   |
| 40385 | COL make VAL FIREFLY COL model VAL 5 COL serie... |      |         |          |     |     |     |     |   | 0     |       |   |
| 40386 | COL make VAL M COL model VAL 6 COL series VAL ... |      |         |          |     |     |     |     |   | 1     |       |   |
| 40387 | COL make VAL NAVION COL model VAL NAVION COL s... |      |         |          |     |     |     |     |   | 0     |       |   |
|       |                                                   |      |         |          |     |     |     |     |   |       |       |   |
|       | match_confidence                                  | gold | left_id | right_id | TP  | TN  | FP  | ... | \ |       |       |   |
| 0     | 0.999998                                          | 0    | 10257   | 2641     | 0   | 1   | 0   | ... |   |       |       |   |
| 1     | 0.999994                                          | 0    | 7344    | 586      | 0   | 1   | 0   | ... |   |       |       |   |
| 2     | 0.999996                                          | 0    | 165     | 2223     | 0   | 1   | 0   | ... |   |       |       |   |
| 3     | 0.999995                                          | 0    | 7824    | 876      | 0   | 1   | 0   | ... |   |       |       |   |
| 4     | 0.999997                                          | 0    | 1318    | 345      | 0   | 1   | 0   | ... |   |       |       |   |
| ...   | ...                                               | ...  | ...     | ...      | ... | ... | ... | ... |   |       |       |   |
| 40383 | 0.999870                                          | 1    | 7173    | 2438     | 1   | 0   | 0   | ... |   |       |       |   |
| 40384 | 0.999997                                          | 0    | 1129    | 256      | 0   | 1   | 0   | ... |   |       |       |   |
| 40385 | 0.999996                                          | 0    | 23805   | 309      | 0   | 1   | 0   | ... |   |       |       |   |
| 40386 | 0.999911                                          | 1    | 6732    | 2316     | 1   | 0   | 0   | ... |   |       |       |   |
| 40387 | 0.999996                                          | 0    | 5816    | 2420     | 0   | 1   | 0   | ... |   |       |       |   |

|       |                                                   | left_agg | TP_agg | TN_agg | \ |
|-------|---------------------------------------------------|----------|--------|--------|---|
| 0     | COL make VAL GULFSTREAM COL model VAL GVI COL ... |          | 2      | 22     |   |
| 1     | COL make VAL PIAGGIO COL model VAL P136 COL se... |          | 1      | 13     |   |
| 2     | COL make VAL AERONCA COL model VAL L3 COL seri... |          | 1      | 8      |   |
| 3     | COL make VAL PZL OKECIE COL model VAL PZL104 C... |          | 1      | 13     |   |
| 4     | COL make VAL BEECH COL model VAL 99 COL series... |          | 1      | 11     |   |
| ...   | ...                                               | ...      | ...    | ...    |   |
| 40383 | COL make VAL NORTH AMERICAN COL model VAL B25 ... |          | 1      | 10     |   |
| 40384 | COL make VAL BEECH COL model VAL 17 COL series... |          | 1      | 14     |   |
| 40385 | COL make VAL CURTISS WRIGHT COL model VAL 22 C... |          | 1      | 15     |   |
| 40386 | COL make VAL MAULE COL model VAL M6 COL series... |          | 1      | 7      |   |
| 40387 | COL make VAL HILLER COL model VAL UH12 COL ser... |          | 1      | 9      |   |

|       | FP_agg | FN_agg |                                                   | right_agg | \ |
|-------|--------|--------|---------------------------------------------------|-----------|---|
| 0     | 0      | 0      | [COL make VAL PA COL model VAL 46 COL series V... |           |   |
| 1     | 0      | 0      | [COL make VAL BHT COL model VAL UH1E COL serie... |           |   |
| 2     | 0      | 0      | [COL make VAL LR COL model VAL 45 COL series V... |           |   |
| 3     | 0      | 0      | [COL make VAL B COL model VAL 737 COL series V... |           |   |
| 4     | 0      | 0      | [COL make VAL BE COL model VAL 18 COL series V... |           |   |
| ...   | ...    | ...    | ...                                               | ...       |   |
| 40383 | 0      | 0      | [COL make VAL BHT COL model VAL 210 COL series... |           |   |
| 40384 | 0      | 0      | [COL make VAL DGA COL model VAL 15 COL series ... |           |   |
| 40385 | 0      | 0      | [COL make VAL CV COL model VAL 340 COL series ... |           |   |
| 40386 | 0      | 0      | [COL make VAL B COL model VAL 737 COL series V... |           |   |
| 40387 | 1      | 0      | [COL make VAL L COL model VAL 1049 COL series ... |           |   |

|       | right_len | match_confidence_max | max_conf_match | best |
|-------|-----------|----------------------|----------------|------|
| 0     | 24        | 0.999907             | 0.999907       | 0    |
| 1     | 14        | 0.999914             | 0.999914       | 0    |
| 2     | 9         | 0.999888             | 0.999888       | 0    |
| 3     | 14        | 0.999797             | 0.999797       | 0    |
| 4     | 12        | 0.999904             | 0.999904       | 0    |
| ...   | ...       | ...                  | ...            | ...  |
| 40383 | 11        | 0.999870             | 0.999870       | 1    |
| 40384 | 15        | 0.999915             | 0.999915       | 0    |
| 40385 | 16        | 0.999895             | 0.999895       | 0    |
| 40386 | 8         | 0.999911             | 0.999911       | 1    |
| 40387 | 11        | 0.999903             | 0.999903       | 0    |

[40388 rows x 21 columns]

```
[194]: df_all_merged[df_all_merged["best"]==1]
```

```
[194]:
```

|   |                                                   | left | \ |
|---|---------------------------------------------------|------|---|
| 6 | COL make VAL BOEING COL model VAL 737 COL seri... |      |   |
| 7 | COL make VAL PIAGGIO COL model VAL P136 COL se... |      |   |

|       |                                                   |
|-------|---------------------------------------------------|
| 19    | COL make VAL CESSNA COL model VAL 172 COL seri... |
| 27    | COL make VAL PILATUS COL model VAL PC6 COL ser... |
| 35    | COL make VAL PIPER COL model VAL L18 COL serie... |
| ...   | ...                                               |
| 40361 | COL make VAL BOEING COL model VAL 737 COL seri... |
| 40374 | COL make VAL AGUSTA COL model VAL AW139 COL se... |
| 40379 | COL make VAL COMMAND AIRE COL model VAL 5C3 CO... |
| 40383 | COL make VAL NORTH AMERICAN COL model VAL B25 ... |
| 40386 | COL make VAL MAULE COL model VAL M6 COL series... |

|       |                                                   | right | match | \   |
|-------|---------------------------------------------------|-------|-------|-----|
| 6     | COL make VAL B COL model VAL 737 COL series VA... |       | 1     |     |
| 7     | COL make VAL P COL model VAL 136 COL series VA... |       | 1     |     |
| 19    | COL make VAL CE COL model VAL 172 COL series V... |       | 1     |     |
| 27    | COL make VAL PC COL model VAL 6 COL series VAL... |       | 1     |     |
| 35    | COL make VAL PA COL model VAL 19 COL series VA... |       | 1     |     |
| ...   | ...                                               | ...   | ...   | ... |
| 40361 | COL make VAL B COL model VAL 737 COL series VA... |       | 1     |     |
| 40374 | COL make VAL AGUSTA COL model VAL AW139 COL se... |       | 1     |     |
| 40379 | COL make VAL COAIRE COL model VAL 5C COL serie... |       | 1     |     |
| 40383 | COL model VAL B25 COL series VAL J COL cert VA... |       | 1     |     |
| 40386 | COL make VAL M COL model VAL 6 COL series VAL ... |       | 1     |     |

|       | match_confidence | gold | left_id | right_id | TP  | TN  | FP  | ... | \   |
|-------|------------------|------|---------|----------|-----|-----|-----|-----|-----|
| 6     | 0.999886         | 1    | 2140    | 895      | 1   | 0   | 0   | ... |     |
| 7     | 0.999916         | 1    | 7345    | 2476     | 1   | 0   | 0   | ... |     |
| 19    | 0.999901         | 1    | 3679    | 1264     | 1   | 0   | 0   | ... |     |
| 27    | 0.999872         | 1    | 11251   | 3243     | 1   | 0   | 0   | ... |     |
| 35    | 0.999812         | 1    | 7478    | 2526     | 1   | 0   | 0   | ... |     |
| ...   | ...              | ...  | ...     | ...      | ... | ... | ... | ... | ... |
| 40361 | 0.999901         | 1    | 1942    | 809      | 1   | 0   | 0   | ... |     |
| 40374 | 0.999902         | 1    | 339     | 158      | 1   | 0   | 0   | ... |     |
| 40379 | 0.943862         | 1    | 21931   | 3620     | 1   | 0   | 0   | ... |     |
| 40383 | 0.999870         | 1    | 7173    | 2438     | 1   | 0   | 0   | ... |     |
| 40386 | 0.999911         | 1    | 6732    | 2316     | 1   | 0   | 0   | ... |     |

|       |                                                   | left_agg | TP_agg | TN_agg | \   |
|-------|---------------------------------------------------|----------|--------|--------|-----|
| 6     | COL make VAL BOEING COL model VAL 737 COL seri... |          | 1      | 7      |     |
| 7     | COL make VAL PIAGGIO COL model VAL P136 COL se... |          | 1      | 13     |     |
| 19    | COL make VAL CESSNA COL model VAL 172 COL seri... |          | 1      | 10     |     |
| 27    | COL make VAL PILATUS COL model VAL PC6 COL ser... |          | 2      | 27     |     |
| 35    | COL make VAL PIPER COL model VAL L18 COL serie... |          | 1      | 7      |     |
| ...   | ...                                               | ...      | ...    | ...    | ... |
| 40361 | COL make VAL BOEING COL model VAL 737 COL seri... |          | 1      | 10     |     |
| 40374 | COL make VAL AGUSTA COL model VAL AW139 COL se... |          | 1      | 12     |     |
| 40379 | COL make VAL COMMAND AIRE COL model VAL 5C3 CO... |          | 1      | 15     |     |
| 40383 | COL make VAL NORTH AMERICAN COL model VAL B25 ... |          | 1      | 10     |     |



|       |                                                   |   |   |
|-------|---------------------------------------------------|---|---|
| 40386 | COL make VAL MAULE COL model VAL M6 COL series... | 1 | 7 |
|-------|---------------------------------------------------|---|---|

  

|       | FP_agg | FN_agg | right_agg \                                       |
|-------|--------|--------|---------------------------------------------------|
| 6     | 0      | 0      | [COL make VAL B COL model VAL 737 COL series V... |
| 7     | 0      | 0      | [COL make VAL P COL model VAL 136 COL series V... |
| 19    | 0      | 0      | [COL make VAL CE COL model VAL 172 COL series ... |
| 27    | 0      | 0      | [COL make VAL PC COL model VAL 6 COL series VA... |
| 35    | 0      | 0      | [COL make VAL PA COL model VAL 19 COL series V... |
| ...   | ...    | ...    | ...                                               |
| 40361 | 0      | 0      | [COL make VAL VARGA COL model VAL 2150 COL ser... |
| 40374 | 0      | 0      | [COL make VAL TCRAFT COL model VAL 22 COL seri... |
| 40379 | 0      | 0      | [COL make VAL ENSTRM COL model VAL F28 COL ser... |
| 40383 | 0      | 0      | [COL make VAL BHT COL model VAL 210 COL series... |
| 40386 | 0      | 0      | [COL make VAL B COL model VAL 737 COL series V... |

|       | right_len | match_confidence_max | max_conf_match | best |
|-------|-----------|----------------------|----------------|------|
| 6     | 8         | 0.999886             | 0.999886       | 1    |
| 7     | 14        | 0.999916             | 0.999916       | 1    |
| 19    | 11        | 0.999901             | 0.999901       | 1    |
| 27    | 29        | 0.999872             | 0.999872       | 1    |
| 35    | 8         | 0.999812             | 0.999812       | 1    |
| ...   | ...       | ...                  | ...            | ...  |
| 40361 | 11        | 0.999901             | 0.999901       | 1    |
| 40374 | 13        | 0.999902             | 0.999902       | 1    |
| 40379 | 16        | 0.943862             | 0.943862       | 1    |
| 40383 | 11        | 0.999870             | 0.999870       | 1    |
| 40386 | 8         | 0.999911             | 0.999911       | 1    |

[3451 rows x 21 columns]

```
[195]: df_all_merged[df_all_merged["best"] == 1]["left_id"].value_counts()
```

```
[195]: left_id
6732    1
2140    1
7345    1
3679    1
11251   1
..
2069    1
293     1
1997    1
9658    1
8169    1
Name: count, Length: 3451, dtype: int64
```

```
[196]: df_all_merged["left_id"].value_counts()
```

```
[196]: left_id
      5382    60
      1512    50
      1514    48
      6818    43
      7552    43
      ..
      3808     3
      1635     3
      8271     3
      2129     3
      1868     2
      Name: count, Length: 3456, dtype: int64
```